

Here is a cleaner, highly structured, and easy-to-read text version designed like a study cheat sheet so you can screenshot or copy it easily for your group:



UTME CHEMISTRY REVISION: SEPARATION TECHNIQUES

1. Filtration

- **Principle:** Separates an **insoluble solid** from a **liquid** based on particle size.
- **Key Terms:** **Residue** (solid left on paper) and **Filtrate** (liquid that passes through).
- **UTME Example:** Sand and water.

2. Evaporation to Dryness

- **Principle:** Recovers a **soluble solid** from a solution by vaporizing the solvent.
- **Condition:** Only used if the solute is **thermally stable** (does not decompose on heating).
- **UTME Example:** Sodium chloride (NaCl) solution.

3. Crystallization & Fractional Crystallization

- **Crystallization:** Used for soluble solids that **decompose easily on heating** (cooled slowly from a saturated solution).
- **Fractional Crystallization:** Separates **two or more soluble solids** based on differences in their solubility at different temperatures.

4. Magnetic Separation

- **Principle:** Uses a magnet to separate **magnetic components** from non-magnetic ones.
- **UTME Example:** Iron filings mixed with sulfur powder or sand.

5. Sublimation

- **Principle:** Separates substances that change **directly from solid to gas** on heating without melting.
- **Must-Know Sublimers:** Iodine, Ammonium chloride (NH₄Cl), Camphor, Benzoic acid, Dry ice.
- **UTME Example:** Separating NH₄Cl from NaCl.

6. Chromatography

- **Principle:** Based on **different rates of migration** of components through a stationary phase carried by a mobile phase.
- **UTME Applications:** Separating plant pigments (chlorophyll), colors in ink, and testing drug purity.

7. Sieving

- **Principle:** Separates solid particles of **significantly different sizes** using a mesh or net.
- **UTME Example:** Separating stones from grains or coarse particles from fine powder.

8. Simple & Fractional Distillation

- **Simple Distillation:** Recovers the **solvent** from a solution or separates a liquid from a non-volatile solid based on boiling points followed by condensation.
- **Fractional Distillation:** Separates **miscible liquids** with **close boiling points** using a fractionating column.
- **UTME Examples:** Ethanol and water mixture, separation of crude oil fractions.

9. Centrifugation

- **Principle:** Uses high-speed rotation to force dense, suspended insoluble particles to the bottom.
- **UTME Example:** Separation of cream from milk or blood cells from blood plasma.

10. Flootation (Levigation)

- **Principle:** Separates a mixture of solids where one component is light enough to **float** on a liquid surface while the heavier one sinks.
- **UTME Example:** Separating chaff from grains.

11. Use of a Separating Funnel

- **Principle:** Separates **immiscible liquids** (liquids that do not mix and form separate layers due to density differences).
- **UTME Example:** Oil and water, or kerosene and water.

 **Golden Rule for UTME:** Always match the technique to the specific **physical property** (e.g., boiling point, solubility, magnetism, density, or state of matter) when answering questions!